

Application Module^X Customer Release Guide

**Release 300
AX04-300**

Application Module^x

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Customer Release Guide**

**Release 300
AX04-300
11/99**

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Honeywell
Industrial Automation and Control
Automation College
2820 West Kelton Lane
Phoenix, AZ 85053-3095

1-800-852-3211

About This Publication

Purpose

The Application Module` (A`M) Customer Release Guide provides

- . An overview of A`M functionality
- . An overview of the functionality that comes with the A`M Release 300
- . Revisions of TotalPlant Solution (TPS) software with which it will work
- . Minimum hardware and firmware revisions and minimum memory sizes for related circuit boards
- . Installation and upgrade preparation instructions
- . Instructions for installing a new A`M
- . Instructions for upgrading an A`M from HP-UX 9.05 to HP-UX 10.20 by performing a cold install, and
- . Problem reporting procedures, should any problems occur during the installation, migration, or ongoing use of the A`M.

A`M Release 300 is compatible with TPS Network Release 433, 531, or 600, or later versions of these releases.

Revision Notes

The following list provides notes concerning the current revision of this document.

Doc ID	Rev. No.	Date	Notes
AX04300	4	11/08/99	Changes made to Revision 4 of this document have been marked with change bars in the outside margin. Overview of Changes 1. Backup procedure steps modified because the pathnames for various daemon files within the inetd.conf files point to R200 locations. 2. Upgrade procedure steps modified because the A`M pre-loaded disk is missing transition links. 3. Upgrade procedure steps modified because after the upgrade ftp failed to work for any account other than root. 4. Upgrade procedure steps modified to change the owner, group or permission. 5. Added a section on installing HP-UX patches.

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1. Contents of the Release

1.1 Overview

Product description

The Application Module` (A`M) provides a dual processor architecture, with the Application Module (AM) providing a reliable and secure link to the LCN and a Hewlett-Packard PA-RISC coprocessor running the HP-UX operating system.

The coprocessor provides an open environment for the development and execution of custom control applications provided by Honeywell or third-party sources, or applications created by the customer.

The Application Module` can be used in different ways, from a “black box” controller, which comes factory-loaded with one or more applications that start automatically when the A`M is brought up, to a sophisticated multi-user development platform and execution engine for multiple control platforms.

1.2 A^xM Functionality

Functionality

The Application Module^x promotes the consistent open solutions evolution theme, since it is another TPS system module with increased capabilities for additional advanced control and optimization applications.

The heart of the Application Module^x is the tightly coupled integration of workstation technology with the TPS system. It utilizes the standard Application Module as its base functional platform, to provide the integration and compatibility with the current TPS system technology. Additionally, the A^xM provides the operating environment of HP-UX and POSIX-compliant interfaces to further enhance the total operating environment by leveraging the use of state-of-the-art commercial software. The A^xM resides on the TPS Network and has an AM front end. It has all the standard AM alarming, CL (background and foreground), CDS data structures, messaging, Regulatory Control, and other point processing capabilities (input/output connections).

TPS Network release requirement

Although the Application Module^x is a TPS system release-independent device, the TPS system must be running on software release 433, 531 or 600, or later for the A^xM to function properly.

Release 300 Differences

The major change to AxM 300 is the replacement of the 9.05 version of the HP-UX operating system with the 10.20 version, which is year 2000 compliant. The major differences between HP-UX 10.20 and HP-UX 9.05 are listed below.

- . The file system structure has changed, particularly the location and/or function of certain system files and directories. In general, these are:
 - /usr (particularly /usr/bin and /usr/lib)
 - /etc, /bin, /lib
 - /dev
 - /system.
- . The impacts of these file system structure changes are:
 - Embedded paths referencing system files that have changed will have to be located and corrected. (HP-UX 10.20 has links for system files that have changed directories.)

- Many system files cannot be restored to their original HP-UX 9.05 path(s). At least one file must be restored to a different directory (`/usr/adm/inetd.sec` to `/var/adm/inetd.sec`).
- Other files must be saved separately, but not restored. You will need to examine these files for custom code that has been added, and then add the code to the new versions of the files.
- The default shell is now the POSIX shell, replacing the Bourne shell that was the default shell in 9.05. The POSIX shell is now in `/usr/bin/sh`. The POSIX shell is a superset of the Bourne shell, so your `sh` scripts, in most cases, will not need to be modified to take account of the change. There are some exceptions. Refer to the discussion of shells in Section 5 of the *Release Notes for HP-UX 10.20* for more information.
- . The Start-up /Shutdown model has completely changed. `/etc/rc` is no longer used. Any custom code that you added to the scripts in `/etc/rc` will have to be identified and re-implemented in the new structure. See subsection 3.4 of this manual for more information.
- . Ignite-UX is a new tool that can be used to do initial configuration and installation of HP-UX systems. It replaces the existing HP-UX Install program. Ignite-UX can also be used to aid in disaster recovery. A bootable tape that is the mirror image of your disk can be created and restored.
- . The Logical Volume Manager (LVM) is now the preferred method of managing disks. LVM is required for root drives larger than 2 GB. See subsection 3.3 of this manual for more information on LVM, and also chapter 3, Managing Disks Using the Logical Volume Manager (LVM), of the *HP-UX System Administration Tasks* manual on the Hewlett-Packard CD-ROM.
- . New software installation tools
 - “swinstall” replaces “update”
 - “swcopy” replaces “updist”
 - “swlist” can be used to list filesets and versions installed on the system

Changes to Honeywell Software are:

- . Repackaging of all dat tapes for compatibility with “swinstall”
- . Repackaging of contents of two tapes:
 - The A`M HP-UX tape is now two separate tapes:
 - HP-UX 10.20 contains ONLY the Unix operating system.
 - The A`M core tape contains the Honeywell software that used to be on the A`M HP-UX tape, and the license manager software.
 - The license manager tape is no longer distributed. The license manger software is included on the A`M Core tape.
- . The Required and Optional patch tapes are no longer distributed.
- . Y2K and HP-UX 10.20 compliance for all Honeywell-supplied applications
 - PCDE server and Wonderware NetDDE for Unix
 - FileLink
 - DMCIx
 - OpenDDA
 - AxM core software
 - AxM personality

Hardware components

The A`M can be configured in a base form, or with one or more options included. The base configuration and additional options are shown in Table 1-1.

Table 1—1 Base and Optional A`M Hardware

Base A`M Hardware	A`M Hardware Options
K2LCN with 4 Mw memory (R433 only) K2LCN with 8 Mw memory (R500 or R600)	K2LCN with 8 Mw memory HMPU plus 4 Mw memory (existing R4xx systems only) K4LCN with 8 Mw memory (R531 or R600 only) K4LCN with 16 Mw memory (R531 or R600 only)
64 MHz PA-RISC processor (on WSI2 board)	100 MHz PA-RISC processor

Table 1—1 Base and Optional A^xM Hardware

Base A ^x M Hardware	A ^x M Hardware Options
4 GB hard drive	1 GB, and 2 GB (existing systems only) 4 GB hard drive Provision for two hard drives on the hard drive tray (8 GB max. total)
32 MB RAM	64 MB, 128 MB, or 256 MB RAM
5-slot chassis	10-slot chassis

1.3 Software Package

Initial software

If purchasing a new Application Module^x, the HP-UX 10.20 software and A^xM core software is already loaded and delivered on the hard drive within the A^xM. The customer must load the A^xM personality. Additionally, the customer must load OpenDDA and any other optional software.

Software components

The table below lists the software components that make up the A^xM software package, along with the marketing kit number, and the Honeywell part number.

Table 1—2 A^xM Standard Software Components

Component Title	Model No	Part No
HP-UX Operating System 10.20 (DAT tape)	MP-SWAXM1	51153333
A ^x M Core on HP-UX 10.20 (Includes License Manager) (DAT tape)	MP-SWAXM1	51153334
A ^x M personality (DAT tape)	MP-SWAXM1	51153335 (R433) 51153336 (R531) 51153337 (R600)
Application Module (AM) Personalities (Zip disk)(Note 1)	MP-M4AM31	51152453
OpenDDA developer's kit (DAT tape)(Note 2)	MP-SWDDA1	51151515

= Note 1: The Application Module (AM) Personalities are required for external load modules AMCL0662 and XACCES, but are delivered with the TPS system software releases 433, 531, and 600.

= Note 2: OpenDDA can be purchased separately for a remote HP-UX workstation.

Software options

The table below lists the software options available for the A⁺M. Each option includes the license and the media for that option.

Attention

Compilers must be upgraded to HP-UX 10.20 versions. You cannot use the compiler tapes from your HP-UX 9.05 system.

Table 1—3 A⁺M Software Options

Software Description	Model No.	Comments
OpenDDA developer's kit (Note 1)	MP-SWDDA1	No compiler
FORTTRAN compiler only	MP-SWDDA2	License and media
C/ANSI C compiler only	MP-SWDDA3	License and media
C/C++ compiler only	MP-SWDDA4	License and media

= Note: OpenDDA can be purchased separately for a remote HP-UX workstation.

1.4 Standard Hardware and Options

Hardware model numbers

The following table provides a list of available hardware and its respective model number for the Application Module[™]. The hardware listed in this table is part of the minimum A[™]M configuration for new builds.

Table 1—4 A[™]M Hardware Model Numbers

Component Description	Model No.	Comments
A [™] M 64 MHz K2LCN w/4 Mw	MP-AMXK21	Part of the minimum configuration for R433
A [™] M 64 MHz K2LCN w/8 Mw	MP-AMXK23	Part of the minimum configuration for R531/R600
32 MB RAM workstation memory	MP-XL32MB	Part of the minimum configuration
4 GB Hard Drive (Primary)	MP-XLAD09	Part of the minimum configuration
PIN Connector (MAU)	MP-PINC01	Part of the minimum configuration

If you do not have a Universal Station[™] or a workstation with a CD-ROM reader and DAT drive to connect to your A[™]M, and you would like to perform your own system administration and development, we recommend the workstation listed below, or an equivalent workstation.

Table 1—5 Optional HP Workstation

Component Description	Model No	Comments
Optional HP workstation recommended for A [™] M System Administration and Development tasks	Hewlett-Packard B132L or B180L	Entry level HP workstation. CD-ROM drive and DAT drive are optional and should be ordered directly from HP.

In addition, if you have the console furniture configuration (US or U[™]S) to accommodate them, an optional DAT drive and/or CD-ROM drives are available that can be configured on your A[™]M.

The table below contains a list of the HP-UX processor memory options

Table 1—6 Optional HP-UX Processor Memory Options

Component Description	Model No	Comments
32 MB RAM workstation memory	MP-XL32MB	Base minimum configuration
64 MB RAM workstation memory	MP-XL64MB	Optional
128 MB RAM workstation memory	MP-XL128M	Optional
256 MB RAM workstation memory	MP-XL256M	Optional

The table below contains a list of the HP-UX hard disk options (new builds and upgrades).

Table 1—7 HP-UX Hard Disk Options

Component Description	Model No	Comments
4 GB Hard Drive (Primary)	MP-XLAD09	Base minimum configuration
Second 4 GB Hard Drive	MP-XLAD10	Optional

Processor options - R433

The table below lists the base minimum processor and the optional processor configurations with HP-UX that can be used with TPS system software release 433.

NOTE: K4LCN processor boards cannot be used with TPS system Release 433. K4LCN boards are only used with Release 500 and 600.

Table 1—8 HP-UX/LCN Processor Options - R433

Component Description	Model No	Comments
A ^x M 64 MHz K2LCN w/4 Mw	MP-AMXK21	Base minimum configuration for R433
A ^x M 64 MHz K2LCN w/8 Mw	MP-AMXK23	Optional for R433; Base minimum for R531 and R600
A ^x M 64 MHz HMPU plus 4 Mw	MP-AMXH11	Optional
A ^x M 100 MHz K2LCN w/4 Mw	MP-AMXK27	Optional
A ^x M 100 MHz K2LCN w/8 Mw	MP-AMXK29	Optional
A ^x M 100 MHz HMPU w/4 Mw	MP-AMXH12	Optional

Processor and memory options - R531 and R600

The tables below list the base minimum processor, optional processor configurations with HP-UX, and K4LCN processor memory options that can be used with TPS system Release 531 and 600.

Table 1—9 HP-UX/LCN Processor Options - R531 and R600

Component Description	Model No	Comments
A ^x M 64 MHz K2LCN w/8 Mw	MP-AMXK23	Base minimum configuration for R531 and R600; Optional for R433
A ^x M 64 MHz K4LCN without memory	MP-AMXK25	Optional
A ^x M 100 MHz K4LCN without memory	MP-AMXK11	Optional

Table 1—10 K4LCN Processor Memory Options - R531 and R600

Component Description	Model No	Comments
8 Mw Mezzanine board	MP-K4MW08	Minimum for K4LCN
16 Mw Mezzanine board	MP-K4MW16	Optional

1.5 Supporting Documentation

Documentation and its contents

Table 1-11 lists published documents that support Application Module^x.

Table 1—11 Supporting Documentation

Document	Media	Contents/Comments
<i>Application Module^x System Administration</i>	Paper Document or CD-ROM	A ^x M system administration tasks. Basic and supplemental HP-UX system administration tasks
<i>Application Module^x Troubleshooting</i>	Paper Document or CD-ROM	TPS system A ^x M problem isolation information
<i>Application Module^x Service</i>	Paper Document or CD-ROM	Configuration and service information
<i>Application Module^x User Guide</i>	Paper Document or CD-ROM	General information on the use of the A ^x M
<i>Application Module^x Customer Release Guide</i>	Paper Document or CD-ROM	Contents of the software release, standard and optional hardware, special considerations when installing A ^x M, and problem reporting procedures
<i>CL/AM Reference Manual, Appendix I</i>	Paper Document or CD-ROM	Use of the AMCL06 external load module
<i>Honeywell License Installation and Administration A^xM</i>	Paper Document or CD-ROM	Describes installing software licenses for A ^x M and OpenDDA
<i>OpenDDA User's Guide</i>	Paper Document or CD-ROM	Information on the use of OpenDDA
<i>OpenDDA Reference Manual</i>	Paper Document or CD-ROM	Detailed reference information on OpenDDA
<i>OpenDDA Customer Release Guide</i>	Paper Document or CD-ROM	How to install and set up the configuration on OpenDDA, porting ABE applications, problem reporting
<i>Hewlett-Packard documentation</i>	CD-ROM or http://www.docs.hp.com	Multiple Hewlett-Packard manuals, covering editors, compilers, networking software, and other tools

2. Special Considerations

2.1 Details

Description of this section

This section is designed to make the customer aware of special situations involving the A`M, the circumstances of those situations, and any actions that can be taken to avoid, or eliminate, each situation.

Be sure to read this section carefully, and take action, as appropriate.

Reduced available memory

Available user memory is reduced for an A`M node as opposed to an AM node of comparable memory size with the same configuration by approximately 1,437,000 words for TPS system Release 433, and by 2,510,000 words for TPS system Releases 531 and 600. This provides space for the increased functionality needed by A`M.

A`M node startup

At least one background CL task or one concurrent data access task must be configured in the NCF for successful A`M node startup.

No A`M redundancy

The A`M nodes cannot be arranged in redundant pairs.

Use of yellow pages reduces performance

Honeywell does not recommend running yellow pages (NIS) on A`M control nodes. Doing so may negatively impact data access performance.

Loading A`M with WSI2 interface off

When attempting to load an A`M with the WSI2 interface present, but with the SHUTDOWN switch in the SHUTDOWN position, the A`M will load. However, it will only load the standard AM personality. To load as an A`M, donot place the SHUTDOWN switch in the SHUTDOWN position.

A`M loading error with licensing

If a WARNING status on the node status display occurs as a result of loading an A`M, there may be a licensing problem. You can see the error by entering the command:

```
more /var/opt/TDC_Open/common/LCN_daemon.log
```

If no error is displayed, enter the following command to check for a licensing problem.

```
more /var/opt/TDC_Open/common/old.LCN_daemon.log
```

See the *Honeywell License Installation and Administration A`M* manual for diagnostic troubleshooting help.

3. Installation or Upgrade Preparation

3.1 Important Preliminary Reading

Overview

If you are installing a new A`M, there is some Honeywell documentation and some Hewlett-Packard documentation that you should read prior to installing your A`M and bringing it on-line. The documentation that is most important for your individual needs depends on what your job function will be with respect to the A`M. If you are upgrading from HP-UX 9.05 to 10.20, there are significant differences in 10.20. The upgrade process is complex and will differ from installation to installation, depending on the amount of customization that you have implemented. For this reason, it is impossible to present a “cook book” step-by-step procedure for performing the upgrade. It is essential that you understand the functionality changes in HP-UX 10.20 in order for the upgrade to be successful. We strongly advise that you read the recommended HP documents as well as the overviews presented in this section.

Honeywell documentation

If your A`M is a new installation with one or more of the functions listed in Table 3-1, we recommend that you read the applicable Honeywell documents prior to installing the A`M.

Table 3—1 Source Information for A`M Functions

Function	Be familiar with the contents of the following Honeywell documents...
LCN or A`M Implementor	<i>CL/AM Reference Manual</i>
System Administrator	All A`M documentation
Maintenance/Technical Support	<i>Application Module^x Service</i> <i>Application Module^x Troubleshooting</i> <i>Honeywell License Installation and Administration A`M</i>
Application Module ^x user	<i>Application Module^x User Guide</i>

Hewlett-Packard documentation

We recommend that anyone who will be using the HP-UX operating system as an operator, developer, or administrator read the appropriate HP documents listed in Table 3-2. It is especially important to read these documents when upgrading your system from HP-UX 9.05 to 10.20 as the migration from 9.05 to 10.20 is not a simple upgrade procedure. They are available on the HP CD-ROM or at the web site <http://www.docs.hp.com>.

Table 3—2 Recommended HP-UX Documentation

HP Document	Importance
<i>HP-UX System Administration Tasks</i>	Important as a general reference for system administrators. Section 3 is especially important as a source of information about the Logical Volume Manager.
<i>HP-UX 10.0 File System Layout White Paper</i>	Important as a general reference for file system layout. Chapter 3 contains important information about the system start-up and shutdown model and about adding custom subsystem to this model. If you are upgrading from 9.05 to 10.20, this document is especially important because there are major file system layout changes in 10.20.
<i>Release Notes for HP-UX 10.20</i>	General reference for HP-UX 10.20 functionality. See the discussion of shells (the POSIX shell has replaced the Bourne shell as the default shell in 10.20).
<i>HP-UX 9.x/10.x Interoperability Guide</i>	Important if you will have a mixed system (9.05 and 10.20) for a period of time.
<i>Upgrading from HP-UX 9.x to 10.x</i>	This document describes the procedure to upgrade from 9.x to 10.01. Another step would be required to upgrade from 10.01 to 10.20. Honeywell has chosen to do the upgrade by doing a "cold install" of 10.20, eliminating the intermediate step. Nevertheless, this document contains a great deal of useful information and should be read before upgrading from 9.05 to 10.20.

3.2 Licensing

Setting up licensing for Honeywell software

If your A`M is new, you must set up the software license prior to operation of your A`M or OpenDDA software. The following steps summarize the software license set up, which is detailed in Honeywell's *License Installation and Administration A`M* manual. The Subsection Number column in this table references subsections in that manual.

= Note: The License Manager software is included on the A`M core software tape and does not have to be loaded separately.

Table 3—3 Honeywell License Setup

Step	Action	Subsection Number
1	OBTAIN the permanent license file.	3.1
2	EDIT the license file.	3.2
3	SET UP the license environment variable.	3.2
4	DETERMINE if an options file is required. IF YES, Set it up.	4
5	START the license servers.	5.1

3.3 Logical Volume Manager (LVM)

Overview

LVM is a disk management subsystem that lets you allocate disk space according to the specific or projected sizes of your file systems or raw data. Logical volumes are the preferred method of managing your disks, and are required in certain configurations.

LVM logical volumes provide capabilities that are not possible when using fixed partitions:

- More flexible and efficient ways to organize your disk(s)
- Supports disk striping which can increase I/O performance
- Supports mirroring which can increase data reliability and system availability (In order to use mirroring, you will need to purchase MirrorDisk/UX, HP product number B2491A)

For more information

If you are planning to use LVM, or need to decide whether or not to use it, see chapter 3, *Managing Disks Using the Logical Volume Manager (LVM)*, of the *HP-UX System Administration Tasks* manual on the Hewlett-Packard CD-ROM or on their documentation web page, <http://www.docs.hp.com>.

When is LVM required?

LVM is required if your root disk is greater than 2 GB because the root partition cannot exceed 2 GB. If you are upgrading to HP-UX 10.20 by purchasing the MP-ZAXMD4 upgrade kit from Honeywell, the kit consists of a 4 GB drive configured for LVM with HP-UX 10.20 and A`M core software preloaded. The following tables show the organization of the 4 GB Drive.

Table 3—4 Logical Volume Assignment for the 4 GB Drive

Logical Volume	Volume Size MBytes	Mount Point
/dev/vg00/lvol1	48	/stand
/dev/vg00/lvol2	128	(swap)
/dev/vg00/lvol3	84	/
/dev/vg00/lvol4	32	/tmp
/dev/vg00/lvol5	20	/home
/dev/vg00/lvol6	236	/opt
/dev/vg00/lvol7	52	/usr
/dev/vg00/lvol8	332	/var
/dev/vg00/lvol9	204	/users

This configuration assigns approximately 25% of the disk space, and the remaining space is available for additional volumes or for expansion of existing volumes. In addition, there is considerable unused space available within the existing volumes.

What if your primary (root) drive is 2 GB or less?

In this case, the use of LVM is optional. However, if your drive is 1 GB or less, you must use whole disk instead of LVM. This is because the total size of the Logical Volume Assignments is very close to 1 GB. You should study Chapter 3 of *HP-UX System Administration Tasks* to determine whether or not you want to use LVM. You will be given the option of selecting LVM or a single-partition non-LVM configuration when you install HP-UX 10.20 on your primary drive. If you choose the non-LVM configuration, it is possible to convert to LVM at a later time. However, it is not a simple procedure because LVM must be configured before loading the operating system. The procedure to convert a root disk to LVM is covered in Chapter 3 of *HP-UX System Administration Tasks*.

What if you have a second hard drive?

If you have a second hard drive, the use of LVM on the second drive is optional. You can convert the second drive to LVM at any time. LVM and non-LVM drives can coexist on your system, but a given physical drive must be either all LVM or all non-LVM. The procedure to convert a non-root disk to LVM is also covered in Chapter 3 of *HP-UX System Administration Tasks*.

3.4 HP-UX 10.20 System Start-up and Shutdown Overview

Overview

The method of starting and stopping subsystems is radically different in HP-UX 10.20 than it was in HP-UX 9.05. This subsection contains a brief overview of the differences, as well as an overview of the new (10.20) model. Developers and administrators who will be writing start-up and shutdown scripts should reference the more comprehensive write-up in Chapter 3 of the *HP-UX 10.0 File System Layout White Paper* for additional information.

Startup HP-UX 9.05 vs. 10.20

In HP-UX 9.05, initialization and start-up of subsystems was performed by `/etc/rc`— either directly executed from `/etc/rc` or by other scripts called from `/etc/rc`. In HP-UX 10.20, the start-up/shutdown architecture separates the execution scripts from the configuration information required for execution. The file structure used to implement this model has three key elements, as shown in the following table.

Table 3—5HP-UX Start-up/Shutdown Key Files

File	Function
<code>/sbin/init.d</code>	This directory contains a pool of execution scripts
<code>/etc/rc/config.d</code>	This directory contains configuration files. Each configuration file corresponds to an execution script in the <code>/sbin/init.d</code> directory, and it is named the same as the execution script.
<code>/sbin/rc#.d</code> where # = <i>run level</i>	These are the sequence directories. Each directory corresponds to a run level. These directories contain symbolic links that point to the execution scripts in <code>/sbin/init.d</code> .

All subsystems are started and stopped from a pool of scripts located in `/sbin/init.d`. The scripts get input from configuration files located in `/etc/rc.config.d`. The system run level (set in `/etc/inittab`) determines which subsystems are executed. `/sbin/rc#.d` (e.g., `rc1.d`, `rc2.d`, `rc3.d`) are directories that correspond to the run level of the system. For a system in run level 3, all subsystem in `rc1.d`, `rc2.d`, and `rc3.d` will be executed.

The following diagram depicts the HP-UX 10.20 start-up/shutdown model.

	Run Level Directories /sbin/rc#.d	Pool of Execution Scripts /sbin/init.d	Configuration Files /etc/rc.config.d
rc1.d	->K100phd ->S100localmount KK	lp cron tdc3000 filelink	lp cron tdc3000 filelink
rc2.d	->K900nfs.server ->S008net.sd KK	phd KK	phd KK
rc3.d	->S100nfs.server ->S240hwlicense KK		
Rc4.d	KK KK		

The configuration variable files located in `/etc/rc.config.d` are used to set variables to control the execution of the startup script. A control variable is always included to determine whether the subsystem is enabled or disabled. Other variables that may be required to start the subsystem can be included. An example of a configuration file is `/etc/rc.config.d/cron`:

```
#!/sbin/sh
# @(#) $Revision: 72.3 $
# Cron configuration. See cron(1m)
#
# CRON:          Set to 1 to start cron daemon
#
CRON=1
```

The files in the sequencer directories (`/sbin/rc#.d`) are simply links pointing to the execution scripts. They are separated into run level directories, as well as numbered to control the start order within each directory. Also included in the filenames are S (start) or K (kill) labels, which determine if the script runs at start-up or shutdown.

The Execution scripts (`/sbin/init.d`) read the corresponding file in `/etc/rc.config.d` to set the variables necessary for execution of the code.

Sequencer directory

The `/sbin/rc` program is invoked by `init`, and runs the execution scripts when transitioning from one run level to another. `/sbin/rc` goes through all levels between the start and target run level (except run level 0, which is a special case—see the HP white paper). When moving up to a run level, all scripts beginning with S will be executed with a start argument for the subsystem. When moving down to a run level all scripts beginning with K will be executed.

Example of a sequencer link:

```
/sbin/rc3.d/S241tdc3000
```

The naming convention is:

<code>/sbin/rc3.d/</code>	The link is in the directory <code>/sbin/rc3.d</code> which corresponds to run level 3
<code>S</code>	The sequencing type (S is for start and K is for kill)
<code>241</code>	The sequence number
<code>tdc3000</code>	Reference name of the script that is to be executed (normally, this would be the same as the name of the script, but it does not need to be)

Within a run level directory (`/sbin/rc#.d`), scripts will be executed in order of the sequence number assigned in the link. For example, a script with a sequencer number of 100 will be executed before a script with a sequence number of 200 or 300 or 351, etc.

Execution scripts

The execution scripts are always invoked with one of four arguments, which indicates the “mode”:

`start` Bring subsystem up

`start_msg` Message to print upon startup. This appears in the system startup checklist.

`stop` Bring subsystem down.

`stop_msg` Message to print upon shutdown.

The start link `S` and the kill link `K` in `/sbin/rc#.d` point to the same script and the argument is given to the script accordingly.

The execution scripts all have the same format, which allows you to use a template when creating your own startup scripts. `/sbin/init.d/template` is provided for this purpose. The file can be edited to your specific subsystem startup. This template is too long to list here in its entirety. The following is a very condensed version indicating the general structure of the template.

Abridged listing of the execution script template

Attention

This is an abridged listing of the execution script template. Do not use this abridged version to create a script. Use the full template located in /sbin/init.d/template.

```
case $1 in
'start_msg')
    echo "Starting <specific> subsystem";;
'stop_msg')
    echo "Stopping <specific> subsystem";;
'start')
    if [ -f /etc/rc.config ] ; then
        . /etc/rc.config
        if [ $CONTROL_VARIABLE != 1 ] ; then
            rval=2
        else
            commands for starting subsystem
        fi
    ;;
'stop')
    if [ -f /etc/rc.config ] ; then
        . /etc/rc.config
        if [ $CONTROL_VARIABLE != 1 ] ; then
            rval=2
        else
            commands for stopping subsystem
        fi
    ;;
*)
    echo "usage: $0 {start|stop}";;
esac
```

When /sbin/rc runs through the startup sequence, it will execute the scripts first with the start_msg argument, then with the start argument.

Sample procedure outline for creating a start-up/shutdown script

The following is an outline of the steps that might be taken to create a startup/shutdown script for an application called new6product. To see a completed script, list one of the files in `/sbin/init.d`.

Table 3—6 Example of Creating an Execution Script

Step	Action
1	<code>cd /sbin/init.d</code>
2	<code>cp template new_product</code>
3	<code>vi new_product</code>
4	Edit the <code>start_msg</code> case statement as follows: Replace <code><specific></code> with <code>new_product</code>
5	Edit the <code>stop_msg</code> case statement as follows: Replace <code><specific></code> with <code>new_product</code>
6	Edit <code>start</code> case statement as follows: - Change <code>CONTROL_VARIABLE</code> to <code>NEW_PRODUCT</code> - Add command: <code>/opt/TDC_Open/new_product/bin/new_product.start</code> after the line that states: <i># Execute the commands to start your subsystem</i> (Multiple commands or control commands are allowed) - Add <code>set_return</code> after command/commands
7	Edit the <code>stop</code> case statement as follows: - Change <code>CONTROL_VARIABLE</code> to <code>NEW_PRODUCT</code> - Add command: <code>/opt/TDC_Open/new_product/bin/new_product.stop</code> after the line that states: <i># Execute the commands to start your subsystem</i> (Multiple commands or control commands are allowed) - Add <code>set_return</code> after command/commands
8	<code>vi /etc/rc.config.d/new_product</code>
9	Add single line: <code>NEW_PRODUCT=1</code>
10	Create the links as follows: <code>ln -s /sbin/init.d/new_product /sbin/rc3.d/S900new_product</code> <code>ln -s /sbin/init.d/new_product /sbin/rc2.d/K100new_product</code>

3.5 Running the HP "Prepare" Tool

Overview

If you have implemented your own applications, or make use of customized scripts, you will want to use the tape labeled "Upgrade Tools." This tape is part of the upgrade kit and contains the "prepare" tool. This software, which flags possible 10.x incompatibilities, can be run on any 9.05 machine that holds the source code for applications and/or customized scripts. The complete instructions for use of this tool can be found on the Internet at <http://www.docs.hp.com>. This web site is available to everyone and contains HP Technical Documentation. To find this information at this site, select *HP-UX Operating System, HP-UX Release 10.x Documentation, Upgrading from HP-UX 9.x to 10.x, Chapter5:ConvertingCode and Scripts*).

How to use the tool to check text files for 10.x incompatibilities

This tool should be run on source code, scripts, and makefiles to check for:

- Invalid pathnames. Pathnames constructed by variable substitution will not be caught (see the *HP-UX 10.0 File System Layout White Paper*).
- Commands, libraries and system calls that have changed (see *Upgrading from HP-UX 9.x to 10.x*, Chapter 8, Compatibility between 9.x Releases and 10.01). This tool will not catch changes to commands and libraries that have occurred since 10.01. Consult *Release Notes for HP-UX 10.10* for these. Also, not all networking services are covered, and changes to graphics and X libraries are not covered (see *Release Notes for HP-UX 10.10* for this information).
- Device file names that have changed (see "I/O Convergence" in the *Release Notes for HP-UX 10.10*).

It is not necessary to update references to system files that have changed directory locations. HP-UX 10.x contains transition links from their old location to the new location.

Running the “prepare” tool

Table 3—7 Procedure to Run the HP Prepare Tool

Step	Action
1	Login to the A ^x M, which is still running HP-UX 9.05, as root. Use the update command to load the upgrade tools tape.
2	Add /upgrade/bin to PATH variable. (setenv PATH /upgrade/bin:\$PATH)
3	Run prepare once without options by entering the command: prepare This will create the .preparerc file that the tool will require later. Answer y when asked if you want to create the file.
4	Review the .preparerc file that you just created. This file is copiously self-documented. Check the defaults for file types and directories to be included and excluded to ensure that you get the results that you want.
5	Next, run prepare on a directory specified by <i>directory_name</i>. You can run the tool on all files by specifying “/” for <i>directory_name</i>. However, it is better to specify a subtree here (for example, /users), to reduce the amount of output the rest of the process generates. By specifying an appropriate subtree, you will avoid generating output for system and Honeywell directories. Prepare -bvr <i>directory_name</i> where: -b tells prepare to do only a quick check of files to see if they need further analysis and conversion. -v tells prepare to report progress to stdout. -r tells prepare to examine <i>directory_name</i> and the directories under it, recursively. This command examines all of the files in your <i>directory_name</i> tree and sorts the files into four categories, creating a file for each category. Each category file contains a list of filenames as follows: 1. SHELL.g this file contains a list of shell scripts and makefiles 1. SRC.g this file contains a list of source files 1. TEXT.g this file contains a list of text files 1. ELSE.g this file contains a list of unrecognized files Each of the files in these lists will have been subjected to a quick check for potential 10.x changes. Four additional category files are created that contain subsets of the above lists. These subsets contain only those files that look as if they will need at least one change. These lists are contained in the following files: 1. SHELL shell scripts and makefiles that potentially require changes 1. SRC source files that potentially require changes 1. TEXT text files that potentially require changes 1. ELSE unrecognized files that potentially require changes

Table 3—7 Procedure to Run the HP Prepare Tool

Step	Action
6	Edit each of these lists (1.SHELL, 1.SRC, 1.TEXT and 1.ELSE), removing files you don't want prepare to look at, such as system files. If you used "/" for the directory name in the previous step, you will see a large number of system files included in the list. These should be deleted.
7	In this step, you will run prepare on each of the lists of files. NOTE: prepare will duplicate not only your files, but also the original directory tree, under the output directory you specify. If you do not have enough disk space to create copies of the files, omit -m NEWDIR from the commands shown below. Use the following commands to analyze the files in each list, creating new files under NEWDIR. If your AXM is on line, you should issue only one command at a time to keep the load to a minimum. <pre>prepare -V -l+2 -L \$ -S \$ -f 1.SHELL -m NEWDIR prepare -V -l+2 -L \$ -S \$ -f 1.SRC -m NEWDIR prepare -V -l+2 -L \$ -S \$ -f 1.TEXT -m NEWDIR prepare -V -l+2 -L \$ -S \$ -f 1.ELSE -m NEWDIR</pre> where: -V tells prepare to report progress (at the file level) to stdout. -l+2 -L \$ creates a logfile 1.SHELL.log -S \$ creates a summary file 1.SHELL.sum -f specifies the name of the file that contains the list of files to be examined. -m creates a new version of each original file under NEWDIR. The new version has 10.x HP-UX pathnames in place of 9.x pathnames, and contains comments on other potential problems, such changed options to commands or system calls. -m creates NEWDIR if it does not exist; if it does exist, old files are not overwritten by new ones. (Use -M instead of -m if you want old files to be overwritten.)

Table 3—7 Procedure to Run the HP Prepare Tool

Step	Action
8	Examine the new files (those under NEWDIR in this example) and do the remaining work of determining necessary changes. The tool adds comment lines containing "#ana lyzer" to the file, in the location where each potential incompatibility is detected. HP includes the tool "fscanner" for locating these comment lines. See the fscanner (update_aid) manpage for more information. You may find "grep" more convenient to use. The appropriate command is: <pre>find NEWDIR -type f -print xargs grep analyzer</pre> The output will show lines like this... <pre>NEWDIR/users/dlb/cdda/alltestc.c:#analyzer, Line 178: Item "exec", WARNING, see LIBS_DB entry L_165</pre> All OpenDDA programs will contain a line for each usage of "exec". This can be ignored. To look up the meaning of an error message, you will need to consult the text files /upgrade/lib/CMDS_DB and /upgrade/lib/LIBS_DB. Search for the string mentioned in the comment line. Using the example above... <pre>pg /upgrade/lib/LIBS_DB</pre> then search for L_165. The HP upgrade tools also provide fnlookup, which maps 9.x pathnames to 10.x pathnames. You can use the command (substitute an absolute pathname below): <pre>fnlookup -o 9.x_absolute_path.</pre> It will return the 10.x pathname.

4. Installation

4.1 Overview

Scenarios

There are three scenarios addressed in this section:

- Installing a new A`M
- Upgrading HP-UX from 9.05 to 10.20 by installing the 4 GB drive as included in the MP-ZAXMD4 upgrade kit
- Upgrading HP-UX from 9.05 to 10.20 by loading software on your existing primary hard drive (MP-ZAXMTM upgrade kit)

4.2 Installing a New A^xM

Steps

If you are installing a new A^xM node, your A^xM hard drive has HP-UX 10.20 and A^xM core software preloaded by Honeywell. The basic functions that you need to perform are (the appropriate references are indicated):

1. Install the hardware and make the necessary network cable connections for the TPS Process Network (LCN) and the Plant Information Network (PIN).
= Reference: Honeywell *LCN System Installation*
2. Connect a local console terminal and DAT drive to the A^xM.
= Reference for steps 2-7: Honeywell *Application Module² System Administration*, sections 2 and 3
3. Set up hostname, IP address, root password, time zone, and other networking parameters.
4. Set up the necessary configuration files so that the A^xM can communicate with other PIN devices, including a device from which it is convenient to do system administration tasks.
5. Load the A^xM personality.
6. Load additional software such as:
 - Compilers
 - OpenDDA
 - Filelink
 - PCDE
 - Other application software as required
7. Create a new A^xM node on the TPS Network including checkpoint and external load modules.
8. Set up licensing.
= Reference: Honeywell *License Installation and Administration A²M*, and subsection 3.2 of this manual.

4.3 Upgrading to HP-UX 10.20

Overview

Because HP-UX 10.20 is considerably different from HP-UX 9.05, upgrading to 10.20 is not a simple process in which you load the new operating system, restore files and run. See Release 300 Differences in subsection 1.2, A `M Functionality, for a list of the major differences between HP-UX 9.05 and 10.20.

Because of these differences it is not possible to perform a standard upgrade in order to update your system from HP-UX 9.05 to 10.20. In order to do so, you would first need to upgrade the system and its applications to 10.01 before updating to 10.20. However, this section provides instructions for moving directly from HP-UX 9.05 to 10.20 by performing a cold install. Installing the software in this way will completely erase the existing operating system.

There are two scenarios for updating your system to HP-UX 10.20:

- . Install the MP-ZAXMD4 upgrade kit (4 GB hard drive with software)
- . Completely reload your primary hard drive

Since many of the steps in these two scenarios are the same, this procedure will cover both, noting exceptions where a step is to be skipped for a given scenario.

Changes in HP-UX 10.20

The major differences between HP-UX 10.20 and HP-UX 9.05 are listed below.

- . The file system structure has changed, particularly the location and/or function of certain system files and directories. In general, these are: `/usr` (particularly `/usr/bin` and `/usr/lib`), `/etc`, `/bin`, `/lib`, `/dev`, and `/system`. The impacts of this are:
 - Embedded paths referencing system files that have changed will have to be located and corrected. (HP-UX 10.20 has links for system files that have changed directories.)
 - Many system files cannot be restored to their original HP-UX 9.05 path(s). At least one file must be restored to a different directory (`/usr/adm/inetd.sec` to `/var/adm/inetd.sec`). Other files must be saved separately, but not restored. You will need to examine these files for custom code that has been added, and then add the code to the new versions of the files. This section provides general guidelines for doing this, but it is impossible to cover all situations.

- The default shell is now the POSIX shell, replacing the Bourne shell that was the default shell in 9.05. The POSIX shell is now in `/usr/bin/sh`. The POSIX shell is a superset of the Bourne shell, so your `sh` scripts in most cases will not need to be modified to take account of the change. There are some exceptions. Refer to the discussion of shells in Section 5 of the *Release Notes for HP-UX 10.20* for more information.
- The Start-up /Shutdown model has completely changed. `/etc/rc` is no longer used. Any custom code that you added to the scripts in `/etc/rc` will have to be identified and reimplemented in the new structure. See subsection 3.4 of this manual for more information.
- Ignite-UX is a new tool that can be used to do initial configuration and installation of HP-UX systems. It replaces the existing HP-UX Install program. Ignite-UX can also be used to aid in disaster recovery.
- The Logical Volume Manager (LVM) is now the preferred method of managing disks. LVM is required for root drives larger than 2 GB. See subsection 3.3 of this manual for more information on LVM, and also chapter 3, Managing Disks Using the Logical Volume Manager (LVM), of the *HP-UX System Administration Tasks* manual on the Hewlett-Packard CD-ROM.

HP-UX 10.20 Hard Disk Requirement

The absolute minimum hard disk size requirement when upgrading to HP-UX 10.20 is 1 GB. However, we strongly recommend that your hard disk is 2 GB or more. See Estimating Hard Disk Requirements in the Options section of the *Application Module²* and *OpenDDA Specification and Technical Data* publication for more information on hard disk requirements.

Backup strategy

The upgrade procedure that follows includes the creation of three backups:

1. Backup number 1 is a full backup of your primary drive that can be used to recover the system to HP-UX 9.05 in case you encounter difficulties with the upgrade. This backup is absolutely essential if you are upgrading your existing hard drive. The upgrade to HP-UX 10.20 will result in the complete loss of all data on your primary hard drive, and therefore this backup tape will provide the only means of recovering your system should the upgrade fail for any reason. SAM is the preferred method for creating this backup.

IMPORTANT: If you do have problems with the upgrade and need to restore HP-UX 9.05 to the hard drive from a local DAT drive, you must have the HP-UX install tape. However, if you are restoring HP-UX from a remote device that has a DAT drive, you will not need the HP-UX install tape. The Backups and Restores section of the *Application Module² System Administration* manual has procedures for restoring files to the hard drive.

2. Backup number 2 is a backup of the configuration and data files that can be restored to the original directories after the upgrade. It is much quicker to restore these files from this backup than to recover selected files from your complete backup.
3. Backup number 3 includes files that you will need for reference and analysis, but that CANNOT BE RESTORED TO THEIR ORIGINAL DIRECTORIES.

Attention

Make sure you have read Section 3, Installation or Upgrade Preparation, of this manual before continuing with this procedure as there are significant differences between 9.05 and 10.20. The upgrade is complex and will differ from installation to installation depending on the amount of customization that you have implemented.

Preliminary information

Later on in the upgrade procedure, you will be prompted for your network information. Before you begin the backup and upgrade procedures, use SAM to determine the following information:

- . Filesystem and device swap space
- . Time zone and current time and date
- . System hostname and Internet Protocol (IP) address
- . Subnet mask
- . Default Network Gateway
 - Gateway hostname
 - Gateway network (IP) address
- . If using DNS, the domain name server
- . Save cron information out to a file: `crontab -l > /etc/crontab.save`

Recommendations

Honeywell strongly recommends that you shutdown your A`M and completely restart it and all your applications BEFORE you begin the upgrade. Sometimes changes are made on-line that are lost after a reboot. Make sure your applications can be restarted on HP-UX 9.05. Begin the upgrade from a known state.

Backup procedure

Attention

The backup procedure is a very important part of the installation procedure. To increase your chances of creating a good backup tape, we recommend that you use a DAT cleaning cartridge to thoroughly clean your DAT drive before creating backup tapes. We also recommend that you make AT LEAST two copies of each of your backup tapes.

Table 4—1 Backup Procedure Steps

Step	Action
1	Run prepare. If you have implemented your own applications, or make use of customized scripts, you should perform a preliminary analysis using the HP “prepare” tool. See subsection 3.5 of this manual. If you plan to use the “prepare”, you should do it now, before proceeding to subsequent steps.
2	Verify that all the links are in place by entering the following command: <pre>fsck -p</pre> Enter y when you see the following prompt: <pre>fsck: /dev/dsk/c201d6s0: root file system continue (y/n)?</pre> If there are errors you will see screen output similar to the following: <pre>/dev/dsk/c201d6s0: FREE INODE COUNT WRONG IN SUPERBLK (FIXED) /dev/dsk/c201d6s0: 8 BLK(S) MISSING /dev/dsk/c201d6s0: BAD CYLINDER GROUPS (FIXED) /dev/dsk/c201d6s0: 16829 files, 0 icon, 357933 used, 51237 free (2965 frags, 6034 blocks) ***** FILE SYSTEM WAS MODIFIED ***** ***** REBOOT HP-UX; DO NOT SYNC (USE reboot -n) *****</pre> Reboot the system if indicated.
3	Invoke SAM.
4	Go to Single User Mode by selecting Routine Tasks, tab to System Shutdown > Go to Single User State and return.
5	Advance to Immediate shutdown using the Tab key. Choose OK and press return.
6	Mount the drive by entering the following command: <pre>#mount -a</pre> NOTE: The bdf command will let you know what is mounted.
7	Create backup number 1, which is a full backup of your primary drive. You must be in Single User Mode to create this backup. See the “A ^x M Backup with local dat” procedure in the Backups and Restores section of the <i>Application Module^x 200 System Administration</i> manual.

Table 4—1 Backup Procedure Steps

Step	Action
8	Verify that backup tape does not have errors by entering the following command: <pre>frecover -x -N -f /dev/rmt/0m -m >& /tmp/chktape</pre> This step may take some time. If the tape does not have errors, the following messages will appear in the file /tmp/chktape: <pre>: file marker read : checkpoint record read</pre> Otherwise, you will receive an error message, and you should create a new tape.
9	Repeat the steps for creating Backup Number 1 to create at least two separate Backup Number 1 tapes.
10	Create backup number 2. Only the directories and files listed below should be part of this backup. However, not all of these files may be present on your system. <pre>/.rhosts /.vue /.vueprofile /etc/group /etc/hosts /etc/hosts.equiv /etc/crontab.save /opt/TDC_Open/netdde/netdde.ini /etc/opt/TDC_Open/common/filelink.cfg (if File Transfer was loaded) /etc/passwd /opt/hwiaclicense/etc/license.dat /etc/nsswitch.conf /etc/resolv.conf /users /etc/opt/TDC_Open/common/xacces.cfg /etc/opt/TDC_Open/common/personality.config Point.parameter files if PCDE was loaded OpenDDA development directories and directories holding executables (/users/axm contains links to applications) Any other user data files</pre>
11	Verify that backup tape does not have errors by entering the following command: <pre>frecover -x -N -f /dev/rmt/0m -m >& /tmp/chktape</pre> This step may take some time. If the tape does not have errors, the following messages will appear in the file /tmp/chktape: <pre>: file marker read : checkpoint record read</pre> Otherwise, you will receive an error message, and you should create a new tape.
12	Repeat the steps for creating Backup Number 2 to create at least two separate Backup Number 2 tapes.

Table 4—1 Backup Procedure Steps

Step	Action
13	Create backup number 3. You can create a tape backup or backup the files to a second drive if you have one. If you do use a tape, use a backup method that backs up files by relative and not absolute pathnames. SAM, which uses <code>fbackup(1M)</code>, is recommended. This backup should include the following files: <pre> /.cshrc /.environ /.exrc /.login /.profile /.session /etc/checklist /etc/conf/dfile /etc/inittab /etc/inetd.conf /etc/netlinkrc /etc/netnfsrc /etc/rc /etc/services /usr/adm/inetd.sec </pre>
14	Verify that backup tape does not have errors by entering the following command: <pre>frecover -x -N -f /dev/rmt/0m -m >& /tmp/chktape</pre> This step may take some time. If the tape does not have errors, the following messages will appear in the file <code>/tmp/chktape</code>: <pre> : file marker read : checkpoint record read </pre> Otherwise, you will receive an error message, and you should create a new tape.
15	Repeat the steps for creating Backup Number 3 to create at least two separate Backup Number 3 tapes.
16	Determine how your 9.05 configuration files have been customized using the "diff" utility to compare the original copies of files located in the <code>newconfig</code> directory with the customized files. For example, to compare the customized files with the original files, enter the following command: <pre>diff /etc/rc /etc/newconfig/rc</pre>

Upgrade procedure

Table 4—2 Upgrade Procedure Steps

STEP	ACTION
1	Connect a Local Console Terminal and DAT Drive. This is covered in subsection 2.3 of the Application Module^x System Administration manual. NOTE: If you are going to use a laptop or other PC with terminal emulation as your console device, it is likely that the terminal emulation software that comes with Windows may not produce readable screen output when used with HP-UX 10.20. See the topic “PC procedure” in the Application Module^x System Administration manual if you experience difficulties.
2	Install the new hard drive. Perform this step only if you are upgrading with the MP-ZAXMD4 upgrade kit (4 GB hard drive with software). Follow the instructions in the kit. After installing the new drive, power up the system. The system will boot and automatically invoke the set_parms script. Configure networking by following the prompts.
3	Load HP-UX 10.20. Skip this step if you are upgrading with the MP-ZAXMD4 upgrade kit (4 GB hard drive with software). This drive contains HP-UX 10.20 and A^xM Core software. Use the procedure in subsection 5.2, Loading HP-UX, of the Application Module^x System Administration manual for A^xM Release 300, to load the HP-UX base software. Make sure you have configured networking as specified in the procedure.
4	Load A^xM Core software. Skip this step if you are upgrading with the MP-ZAXMD4 upgrade kit (4 GB hard drive with software). This drive contains HP-UX 10.20 and A^xM Core software. Use the procedure in subsection 5.4, Installing Other Software, of the Application Module^x System Administration manual for A^xM Release 300 to load the A^xM Core software.
5	Skip this step if you are loading from scratch. Otherwise, perform this step if you are upgrading with the MP-ZAXMD4 upgrade kit (4 GB hard drive with software). Install the transition links: <pre>/opt/upgrade/bin/tinstall</pre>
6	Load A^xM Personality. Use the procedure in subsection 5.4, Installing Other Software, of the Application Module^x System Administration manual for A^xM Release 300 to load the A^xM Personality.
7	Load Compilers and OpenDDA. If you will be using OpenDDA on this A^xM, load the compilers that you will use, followed by OpenDDA software.
8	Load other software. Load any other applications that you will use.

Table 4—2 Upgrade Procedure Steps

STEP	ACTION
9	Using SAM, restore backup number 2. Restore the entire contents of backup number 2 to the original paths. Make sure “Overwrite Newer Files” is checked under “Additional Parameters”. The Posix shell must be the default shell for startup. You have just restored your old <code>/etc/passwd</code> file, which specifies the startup (root) shell (and it probably is not Posix). Therefore, you must change the <code>/etc/passwd</code> file as follows: <pre>chsh root /sbin/sh</pre> ATTENTION: Do not log off or reboot until you have completed all the steps in this procedure table.
10	Add the file <code>/etc/shells</code> and include the following lines: <pre>/sbin/sh /usr/bin/sh /usr/bin/rsh /usr/bin/ksh /usr/bin/rksh /usr/bin/csh /usr/bin/keysh</pre> other shell pathnames that are referenced as startup shells in the <code>/etc/passwd</code> file
11	Alternatively, you can use SAM to modify the user's account: Sam > Accounts for Users and Groups > Users> action > Modify > Start-up Program Then choose from the following list: <pre>/usr/bin/sh /usr/bin/rsh /usr/bin/ksh /usr/bin/rksh /usr/bin/csh /usr/bin/keysh</pre>
12	From the backup number 3 tape, restore the file: inetd.sec from the following directory: <pre>/usr/adm/</pre> To the following directory: <pre>/var/adm/ (use the same file name)</pre> You may see the error “file <code>./adm</code> exists but is not a directory”. You can ignore this error.

Table 4—2 Upgrade Procedure Steps

STEP	ACTION
13	ATTENTION: The steps performed up to this point have been fairly straightforward. However, the remaining steps are potentially more complicated. You will be asked to examine, merge or reconfigure certain files. You should already have determined any customizations to your configuration files when you used the "diff" utility as part of the backup procedure. It is also important that you read either the Installation or Upgrade Preparation section of this manual or the pertinent HP-UX documentation before continuing with these steps.
14	Merge any custom startup code that you had in <code>/etc/rc</code> into the new structure. Refer to subsection 3.4 in this manual, and use your copy of <code>/etc/rc</code> from backup 3 as a reference.
15	Examine the old (from backup 3) version of <code>/etc/services</code> and merge any customization into the new version.
16	If you had file system swap configured prior to the upgrade, and you did not replace it with device swap during the HP-UX 10.20 installation you can use SAM to reconfigure any desired file system swap. See Section 4.4, Adjusting Swap Space, of the <i>Application ModuleX System Administration</i> manual for Release 300.
17	Examine the logins and profiles for your users in <code>/user.s</code> . Check for path variables that may have changed.
18	If you have done extensive customization to your kernel, you can examine your saved copy of <code>/etc/conf/dfile</code> , which contains changes from the defaults. Use SAM to rebuild the new kernel to incorporate your customizations, but you should refer to the HP documentation to verify that the changes comply with HP-UX 10.20.
19	If you ran the prepare tool (step 1), refer to the output generated by that tool to determine if any other changes are required.
20	Verify that the Honeywell Licensing Installation has been performed before rebooting the system. See subsection 3.2, Licensing, of this manual.

Table 4—2 Upgrade Procedure Steps

STEP	ACTION
21	Use chown, chgrp, or chmod commands to change the owner, group, or permission. <code>cd /etc/opt/TDC_Open/common</code> <code>. chown root xdapriority.cfg chgrp bin xdapriority.cfg</code> <code>. chown root xaccess.cfg chgrp bin xaccess.cfg</code> <code>cd /opt</code> <code>. chown root TDC_Open chgrp bin TDC_Open</code> <code>cd /opt/TDC_Open</code> <code>. chown root common chgrp bin common</code> <code>cd /opt/TDC_Open/common</code> <code>. chown root bin chgrp bin bin</code> <code>. chown root newconfig chgrp bin newconfig</code> <code>. chown root share chgrp bin share</code> <code>. chown root lib chgrp bin lib</code> <code>cd /opt/TDC_Open/common/lib/</code> <code>. chown root libcds.sl chgrp bin libcds.sl</code> <code>cd /dev</code> <code>. chgrp other ptyp*</code> <code>. chgrp root ptyp1</code> <code>. chgrp other ptyq*</code> <code>. chgrp other ptyr*</code> <code>. chgrp other ttyp0</code> <code>. chgrp other ttyp3</code> <code>cd /</code> <code>. find /var -type f -print xargs chmod 755</code> <code>. find /var -type d -print xargs chmod 755</code> <code>. chmod 777 /var/tmp</code>

4.4 HP-UX Patches for A^xM

Installation Instructions

All patches for HP-UX 10.20 are installed using the *swinstall* command. If you choose to install both patch bundles on the A^xM, we recommend installing the entire patch bundle for both products from the *U^xS/A^xM HP-UX Patches* tape. Installing both patch bundles on the A^xM will take approximately five hours. Each patch bundle **MUST** be installed separately. The *Extension Software Patch Bundle* must be installed before the *HP-UX Core OS Year 2000 Patch Bundle* is installed.

ATTENTION

Do **NOT** abort the installation before the patches are completely installed. Doing so could corrupt some of the files.

Extension Software Patch Bundle and HP-UX Core OS Year 2000 Patch Bundle for the A^xM

Table 4—3 Installing HP-UX 10.20 Patches on the A ^x M	
Step	Action
1	Back up your system before installing patches.
2	Shutdown the LCN personality.
3	Insert the <i>U^xS/A^xM HP-UX Patches for HP-UX 10.20</i> tape into the DAT tape drive.
4	Log onto the system as root and enter the root password.

Table 4—3 Installing HP-UX 10.20 Patches on the A ³ M	
Step	Action
5	Enter the command: <code>/usr/sbin/swinstall</code> Press Enter.
6	A navigation window appears explaining the use of the console terminal keys. If you are not familiar with the use of these keys, read the instructions. Press Enter to continue.
7	A window titled SD Install - Software Selection (<i>hostname</i>) (1) appears.
8	The install software will scan the default source depot type and will display a Note (<i>hostname</i>) window with a message displayed and with the OK selection highlighted. Press Enter to continue.
9	A Specify Source (<i>hostname</i>) window appears with the entry in the Source Depot Type highlighted. Select local tape (or Network Directory/CDROM) as the source by performing the following: - Press Enter to display additional options, if necessary. - Press the Tab key to go to OK and then press Enter.
10	A Bundles and Products window appears with the two patch bundles on the <i>U^S/A³M HP-UX Patches</i> tape listed.
11	Select Options > Change Options and verify that ONLY the following options have been selected. - Create target path if not already there - Mount filesystems in <code>/etc/fstab</code> or <code>/etc/checklist</code> - Reinstall files even if same one already there - Use checksum when checking if file is the same - Autoselect dependencies when marking software - Enforce dependency analysis errors in agent - Enforce script failures - Enforce disk space analysis errors Click OK after making any necessary corrections to the available options to return to the SD Install - Software Selection (<i>hostname</i>) window.
12	Select the <i>Extension Software Patch Bundle</i> . Mark the patch bundle for installation by performing the following: - Highlight the patch bundle using the space bar. - Press the Tab key to go to the menu bar. - Use the right arrow key to move to the Actions drop down menu and then press Enter. - Use the down arrow key to select Mark for Install from the Actions menu and then press Enter. The patch bundle product should now be marked Yes for Install.

Table 4—3 Installing HP-UX 10.20 Patches on the A[®]M

Step	Action
13	An Error (hostname) window may appear if the system detects any software dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Press Enter to return to the Bundles and Products window.
14	Select Install (Analysis) from the Actions menu and then press Enter.
15	An Install Analysis (hostname) (2) window appears with Target, Status, and Products Scheduled information displayed. The install program is checking for adequate disk space and software dependencies. - If all dependencies have been met and there is adequate disk space, the Status will indicate Ready with errors - You will see the following report in the logfile: 2 of 255 filesets had errors, 63 of 255 will be skipped, 190 of 255 had no errors or warnings. - Press the Tab key to go to OK and then press Enter.
16	A Confirmation (hostname) window appears. Select Yes and press Enter.
17	Some software requires a system reboot after installation. - If your system requires a reboot, another Confirmation (hostname) window will appear indicating that the system will be rebooted when installation is complete. - If this occurs, select Yes and press Enter.
18	The Install Window (hostname) (3) window appears during the installation process displaying Target, Status, Percent Complete, Kbytes Installed, Time Left (minutes) and Loading Software information. - The Percent Complete will advance to 100%. - Do NOT begin the next step until the Status indicates Completed or Ready.
19	Press the Tab key to go to Logfile and then press Enter to display all the loaded products.
20	A Logfile (hostname) (2) window appears. - Scroll through the log to determine if there were any errors during the installation. - Press the Tab key to go to OK and then press Enter.
21	Repeat the procedure, but select the <i>Year 2000 Patch Bundle</i> . After analysis, you will see the following information in the logfile: 11 of 33 have errors, 16 of 33 are skipped, 6 had no errors or warnings. This is an expected result
22	The system will return to the Install Window (hostname) (3) window. Press the Tab key to go to DONE and then press Enter.

Table 4—3 Installing HP-UX 10.20 Patches on the A ^z M	
Step	Action
23	From the SD Install - Software Selection (<i>hostname</i>) (1) window, press the Tab key to go to the menu bar (File will be selected) and then press Enter. Select Exit from the pull-down menu and then press Enter.
24	Type the following at the prompt to reboot the system: /etc/reboot
25	Reload the LCN personality.

Troubleshooting

If the kernel rebuild fails, then you probably did not install the bundles separately. The installation will suspend. Re-invoke *swinstall* and re-install the bundles one at a time. You must use the Change Options menu item, and select items that will allow you to reinstall the same revisions of filesets and files.

Additional A^zM Information

See the following sections in the *A^zM System Administration* manual for additional information on installing non-operating system software:

A^zM System Administration manual

Section	Subsection
Installing and Updating Software	Copying Software to a Depot
	Installing Other Software

5. Problem Reporting Procedures

5.1 Overview

Introduction

When it comes to troubleshooting and successful diagnosis of a migration problem, understanding and recording what occurred before and during a failure is extremely important.

The following procedures should be used for reporting problems encountered during the migration.

5.2 Collect Failure Data

Overview

If a failure has occurred, specific data should be collected that will help in the diagnosis of the problem. With the A^xM, a failure could occur on either the AM-side (CL and LCN-related) or on the X-side (HP-UX, application program-oriented), and either side may still be running while the other has failed. Determine which side has failed before proceeding and obtain the related information as it applies. If you are unsure, collect all of the information listed in this section.

If it's an X-side failure

Enter the following commands to show the version of the X-side software.

Table 5—1 Procedure to Identify Version of X-Side Software

Step	Action
1	ENTER what /opt/DDA/bin/dda sum -p /opt/DDA/lib/*.sl (for OpenDDA)
2	ENTER what /opt/TDC_Open/common/bin/xaccess for the XACCESS security restart tool
3	ENTER what /opt/TDC_Open/common/bin/xdacfg for the xdaconfig priority/channel configuration tool
4	ENTER what /opt/TDC_Open/common/bin/cds_hdr for the CDS header generation tool
5	ENTER what /opt/TDC_Open/common/bin/lcndaemon743 what /opt/TDC_Open/common/bin/cdsdaemon what /opt/TDC_Open/common/lib/lib* Include for all A ^x M problems found.
6	ENTER what /opt/hwiaclicense/bin/hwiacmgrd Include for all licensing problems found.

Also, include the following information:

Table 5—2X-Side Failure—Data Collection

Information (or command)	Description
/usr/adm/syslog	The HP-UX system log file
Core dumps	Core dumps in your current working directory (non-CL-initiated applications)
/users/axm/core	Core dumps (CL-initiated applications)
/var/opt/TDC_Open/common/CDS_daemon.log	Daemon log file
/core	Provide only if the CDS daemon crashes and core dumps
/core	Provide only if the LCN daemon crashes and core dumps
/var/opt/TDC_Open/common/LCN_daemon.log	Audit trail log file
/var/opt/TDC_Open/common/old.LCN_daemon.log	Previous audit trail log file

For OpenDDA subsystem PARs (Product Anomalies Report), please enclose the following additional information:

- A copy (electronic preferred) of the application's source file
- A copy of the application's definition file (.def)
- A screen dump, if using the Command Line Interface (CLI)
- A printout of the application's output file, if the error occurred during execution (.dda6out)

For licensing PARs, please include the following information:

- If the problem occurs during node load, include the `LCN_daemon.log` and the `old.LCN_daemon.log`, from the locations noted in Table 5-2.
- If the problem did not occur during node load, include the `/var/hwiaclicense/log/log` along with the error reported by the licensed software component (for example: `/opt/DDA/bin/dda`).

If using the “XACCESS” security restart utility, collect the following information:

Table 5—3 Failure Data Collection if Using XACCESS

Information (or command)	Description
Core	XACCESS utility core dumps; found in your current working directory.
/etc/opt/TDC_Open/common/xaccess.cfg	The XACCESS security restart file
Any messages printed to the terminal.	

If using the “XDACONFIG” priority/channel configuration tool, collect the following information:

Table 5—4 Failure Data Collection if Using XDACONFIG

Information (or command)	Description
Core	XDACONFIG utility core dumps; found in your current working directory.
/etc/opt/TDC_Open/common/xdacconfig.cfg	The XDACONFIG configuration tool restart file
Any messages printed to the terminal.	

If it's an AM side failure

Include versions for the following: Personality version (from SMCC when the node is loaded)

- . All external load modules that are loaded (.LO)

Then, examine the symptom labels in this section:

- . System or module working, but not properly,
- . Node not responding,

and where applicable, use those procedures to collect failure data.

System or module working, but not properly

Perform the following steps as they apply to your situation.

Table 5—5 Steps to Take if You Have an Unknown Failure

Step	Action	Examples/Hints
1	Record system software release no.	R431.0 for example
2	Record all symptoms of failure.	Slow response from node Error indication on node
3	Record all user actions that occurred immediately before the failure.	Type of operation being performed, such as • Area database configuration • Point building • Building trend displays • Building graphic displays
4	Record the personalities being used.	Universal, engineering, operator, History Module, Application Module
5	Record error messages during the time of the error (in their entirety).	Real Time Journal printouts Event History Retrieval printouts
6	If custom graphic displays or user-written CL programs are involved, copy related information.	Schematic source files Related subpictures Related CL programs
7	If the problem is related to particular points, IDFs, or exception-build files, submit them to TAC.	
8	If you have the TLK1 Tool kit, use NODEPERF to print the screen of data shown for the node.	For use of Tool kit, see the <i>Customer Resource Manual</i> , Section 15.

Node not responding

Perform Steps 1 through 8 in “system or module working, but not properly” earlier in this section, then perform the following steps as they apply to your situation.

Table 5—6 Steps to Take if Node Is Not Responding

Step	Action	Result/Notes
1	Record state of LEDs on module control and processor boards.	
2	Record value in 3-digit display on processor board.	-191 for example.
3	Use SMCC to display Detailed Module Errors for the failed node, and print a copy of the display.	Displays last error in the personality.
4	Dump contents of memory to removable media, using dump function.	See <i>Engineer's Reference Manual</i> , Section 20. Note: Cartridge preferred. (New cartridge will be sent to customer)
5	Copy contents of !CSY directory from NET to removable media, using this command. CP NET>!CSY>*.* \$Fd>!CSY>=-D where d = destination drive	Cartridge preferred. Use same cartridge as in Step 4 of this procedure.
6	Obtain printout of board type and revision status of failed node. Reload failed node with same personality it had when it failed, then • Select SMCC from Engineering Main Menu • Select REV/CONFIG Status • Perform a screen print	
7	Print at least the last two pages of the Real Time Journal that showed the failure.	
8	Print Event History Retrieval for all system errors immediately before and during the time of the failure.	
9	If hardware error is suspected, run appropriate HVTS software.	See <i>Hardware Verification Test System</i> in <i>LCN Service - 2</i> binder.

5.3 Field Support Plan

Field support plan

To support the A`M, the following field support plan is in place.

The field support and problem reporting information in this manual is specifically for customers who are supported by Honeywell service organizations residing in the continental United States. If you are serviced by one of the many Honeywell Regions outside the continental United States, you must contact your local office to obtain the service contact information for your system.

Customers should appoint a liaison for reporting problems and asking questions of the Honeywell Solutions Support Center. A customer focal point minimizes multiple calls to center for the same problem and develops a customer system expert. **Technical Training**

Please refer to the *Automation College Catalog* for complete descriptions, appropriate prerequisites, course dates, and prices.

If you have additional questions regarding technical training issues, please contact your project manager, or the Honeywell Automation College in Phoenix, toll free 1 (800) 852-3211 [in Arizona (602) 313-5669].

Field Problem Resolution

- In the continental United States, the Solutions Support Center group receives all requests for field support. Solutions Support owns and maintains their own TPS system, including all types of nodes such as the A`M. They attempt to duplicate customer-reported problems on this equipment, as well as using it for training and system practice.
- For customers within the continental United States serviced by Honeywell Solutions Support Center in Phoenix, the method of reporting and resolving any problems uncovered in the field is as follows:

The person on-site should contact Solutions Support only, with all suspected system or documentation problems. Be sure to have accumulated all information that should be available before calling for assistance. When calling, be sure to stress the relative seriousness of the problem using appropriate terms (for example, loss of view, emergency, production stopped, etc.).

- On the A`M, TAC access is available through a modem interface. For this interface, we recommend you use a US Robotics Sportster 9600 (or equivalent) modem

Table 5—7Honeywell TAC Support Information

TAC Information	Telephone Numbers/Details
Honeywell TAC phone number (24 hours a day)	1 (800) 822-7673 -or- In Arizona: (602) 313-5558
7:00 A.M. through 4:00 P.M. MST	TAC dispatcher will route your call to the appropriate TAC engineer.
4:00 A.M. through 7:00 P.M. MST	Emergency calls go to an answering service, then an on-call TAC engineer responds to calls of a critical nature.

- TAC gathers enough information to complete a Product Anomalies Report (PAR). TAC clarifies customer operational errors or documentation errors. If the problem is a documentation error, TAC turns in the documentation bug report. They help the customer gather enough data to reproduce and describe the problem.
- TAC then analyzes the problem. If TAC cannot analyze the problem, they work with the customer to gain enough information to identify the source of the problem. TAC and Engineering may attempt remote analysis if the problem cannot be defined in Phoenix. Analysis can be performed on-site if remote diagnosis is unsuccessful and the problem is a critical one.
- TAC gives the US Honeywell Software Release Center the PAR and tracks the problem priority and status for the customer.
- The customer is informed a PAR has been written and TAC communicates to the customer any work-arounds that apply to this problem.
- Engineering analyzes the problem, corrects it if reasonable to do so, retests the problem area, and ensures that the fix introduces no system problems. Engineering then puts the fix into the source, recompiles the source, regression-tests the product (or the system as required), and puts this corrected code into the next system Software Update for field distribution. (Note: The “source“ may be a patch source file.)

- If the problem is critical, modified software may be sent to the site to temporarily remedy the problem. A system Software Update, however, is the distribution technique for the final problem resolution.
- The US Software Release Center is the single distribution point for TPS system LCN Module System Software, which includes the A`M software. This ensures a file of customer names and their software revision exists. Additionally, this ensures that customers receive software updates as they become available.

Spare parts support

For customers within the continental United States, send purchase orders for spare parts to:

Honeywell IAC
Parts Support Center - M/S 455
1100 Virginia Drive
Fort Washington, PA 19034

In an emergency, call 1 (800) 223-8947.

The Houston Depot contains all A`M replacement parts for use by the U.S. region.

If a part fails at a customer site, the customer or Honeywell person calls the local Information Service Center (ISC) branch office. The branch office then arranges for the depot to send a spare to the site within 24 hours.

For emergency parts service after hours or on weekends, call the Houston Depot after-hours number at:

(713) 863-3331

Field Quality group

TPS system hardware, firmware, and software must all be at the correct revision level to guarantee that they function together.

The Field Quality group ensures that all customer systems and spare parts, as well as the Spare Parts Depot, maintain their hardware at the correct revision level for the software release being shipped.

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FOR ADDITIONAL ASSISTANCE:

Write	Call
Honeywell Inc. Industrial Automation and Control Automation College 2820 West Kelton Lane Phoenix, AZ 85053-3028	Technical Assistance Center (TAC) 1-800-822-7673 (48 contiguous states except Arizona) 602-313-5558 (Arizona)

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Industrial Automation and Control

Automation College

2820 W. Kelton Lane

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